

HiMCM 2021 Problem A — Summary Sheet

Team XXXX sources: `annual_pv_load_8760h.csv`, `optimal_sizing_solution.csv`, `budget_pareto_front.csv`,
`lcos_emerging_tech.csv`, `crewai_adaptive_vs_passive.csv`

What we built. Hay–Davies PV and Markov load (8760 h). SOC/SOH pack physics. SciPy HiGHS MILP for $x_k \in \{0..4\}$ and typical-day EMS. Budget Pareto front. Neuro-symbolic crewAI DSR bound to the appliance catalog. LCOS for LFP / Na-ion / VRFB / cement. Figures are script outputs. No hand-edited CSV cells.

Task 1 — Solar and load (`annual_pv_load_8760h.csv`). PV **12,266 kWh/y** (mean 1.400 kW). Load **15,398 kWh/y** (mean 1.758 kW). Energy gap **3,132 kWh**. LPSP cannot fall below $\approx 20\%$ with this array, with or without packs.

Tasks 2–3 — \$10,000 BOM (`optimal_sizing_solution.csv`). **3 FREEDOH packs + 1 lead-carbon pack** (24.6 kWh, 10.5 kW, capex **\$9,700**). Tesla Powerwall+ costs \$10,500. It is **not** in this mix. It appears only at the \$18k Pareto-front row. 8760-hour LPSP **0.319**; outage hours **3,057**; outage events **366**; 10-year TCO \$157,669.

Budget knees (`budget_pareto_front.csv`). We mark \$8,000 and \$10,000 on the Pareto-front figure. \$8k BOM: 2 FREEDOH + 1 lead-carbon, **3,576 h / 391 events**, LPSP 0.362. \$10k: as above, **3,057 h / 366 events**. Events are contiguous deficit runs (hundreds per year). They are *not* “six outages.” \$12k–\$15k is a hardware plateau (same BOM, 2,810 h).

crewAI DSR (`crewai_adaptive_vs_passive.csv`). Days 40–42 (72 h, 2021-02-09 to 02-11) on the \$10k fleet: shed **22.66 kWh**; unserved energy **22.51 $\rightarrow$ 9.22 kWh**; unserved hours **16 $\rightarrow$ 5** (longest run 10 $\rightarrow$ 3 h). Both SOC traces rest on $SOC_{\min} = 0.178$. We do **not** report a 32.5 h blackout cleared to zero or a 9.2% SOC hover. Those values are not in the CSV.

Task 4 — LCOS (`lcos_emerging_tech.csv`, $r = 0.07$, $N = 20$). Sodium-ion **\$0.161/kWh** (lowest; TRL 7 scenario). VRFB \$0.292; LFP baseline \$0.410; cement-structural \$1.273 (TRL 4).

Task 5 — Homeowner takeaway. Do not buy panels as a winter-pack substitute. Size storage from usable kWh and DoD, not nameplate slogans. In dark-cold hours, defer dryer/dishwasher, dim lights, and keep fridge, well, router, and the work laptop. See `letter_to_homeowners.tex`.